# Simple polygons whose sides lie on $n$ lines: a parity obstruction and constructions

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**Abstract**

Let $M(n)$ be the largest number of sides of a simple polygon all of whose sides lie on $n$ lines. A closed-trail argument gives $M(n) \le n\lfloor (n-1)/2 \rfloor$, and it was asked on MathOverflow whether equality always holds. For odd $n = 2r+1$ we prove two things. First, in every simple arrangement of $n$ pseudolines, a certain 2-regular graph on the intersection points, the alternating 2-factor, has a number of components congruent to $(n^2-1)/8 \pmod 2$. Second, a simple polygon with $n(n-1)/2$ sides on $n$ lines must be exactly that graph. Consequently, if $n \equiv \pm 1 \pmod 8$ then $M(n) \le n(n-1)/2 - 3$; in particular the conjectured equality fails for $n = 7, 9, 15, 17, \ldots$. The proof of the parity statement is elementary: a sweep of the wiring diagram and a potential function on permutations. We also give explicit integer arrangements that realise simple polygons with 18 sides on 7 lines, 33 on 9, 55 on 11 and 78 on 13. Hence $M(7) = 18$ and $M(9) = 33$, while $M(11) = 55$ and $M(13) = 78$ attain the bound. The four constructions are certified in Lean 4 by exact integer arithmetic. The parity theorem and the upper bounds are proved on paper only; they are confirmed by exhaustive computation over all simple arrangements of at most 9 pseudolines, but are not formally verified.

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# 1 Introduction

A polygon is *simple* if its boundary is a Jordan curve. Suppose a simple polygon has $k$ sides $S_1, \dots, S_k$ and each side $S_j$ lies on a line $L_j$ taken from a fixed set of $n$ lines. How large can $k$ be? We write $M(n)$ for the maximum, over all sets of $n$ lines and all such polygons.

This question was asked on MathOverflow by Johnathan Will (https://mathoverflow.net/users/596423) on July 31, 2026 [8]. Their upper bound comes from the following observation. Consecutive sides lie on distinct lines, and since the vertex $L_j \cap L_{j+1}$ is visited only once, no unordered pair of lines occurs twice as a consecutive pair. So the cyclic sequence $L_1, \dots, L_k$ is a closed trail in the complete graph $K_n$, and $k$ is at most the maximum length of a closed trail in $K_n$, which is $n\lfloor (n-1)/2 \rfloor$ (OEIS A050187 [11]).

The asker conjectured that $M(n) = n\lfloor (n-1)/2 \rfloor$ for every $n$. They gave simple examples for $n \le 6$, using the pentagram outline for $n = 5$ and the Star of David for $n = 6$, and self-intersecting examples based on star polygons for $n = 7, 9, 10$; the latter construction breaks down at $n = 11$. In the comments, the user DSM (https://mathoverflow.net/users/12613) found a simple 24-gon on 8 lines and remarked that they could reach 18 sides with 7 lines but not 21. The asker then asked whether one can prove that for $n = 7$ only a self-intersecting ("genus 1") polygon can reach 21 sides. As of October 8, 2026 the question had no answers.

**Results.** Write $n = 2r + 1$ for odd $n$; then $n\lfloor (n-1)/2 \rfloor = \binom{n}{2}$.

**Theorem 1.1** (Parity obstruction)**.** *Let $n = 2r + 1$. In every simple arrangement of $n$ lines, and more generally of $n$ pseudolines, the* alternating 2-factor *$G$ (Definition 2.2) has a number of connected components congruent to $r(r+1)/2 = (n^2-1)/8$ modulo 2.*

**Theorem 1.2** (Reduction)**.** *If $n$ is odd and a simple polygon with $\binom{n}{2}$ sides has its sides on $n$ lines, then the arrangement is simple and the polygon is exactly the alternating 2-factor, which must therefore be connected. If instead such a polygon has $k \ne \binom{n}{2}$ sides, then $k \le \binom{n}{2} - 3$.*

**Corollary 1.3.** *If $n \equiv \pm 1 \pmod 8$, then $M(n) \le \binom{n}{2} - 3$. In particular $M(7) \le 18$ and $M(9) \le 33$, so the conjectured equality fails for every $n \equiv \pm 1 \pmod 8$.*

**Theorem 1.4** (Constructions; Lean-certified)**.** *There are explicit arrangements of 7, 9, 11 and 13 lines with integer coefficients carrying simple polygons with 18, 33, 55 and 78 sides respectively (Section 4). Consequently*

$$M(7) = 18, \qquad M(9) = 33, \qquad M(11) = 55 = 11 \cdot 5, \qquad M(13) = 78 = 13 \cdot 6.$$

The values $M(11)$ and $M(13)$ follow from the constructions together with the elementary upper bound. The values $M(7)$ and $M(9)$ also need Corollary 1.3. The value $M(7) = 18$ confirms DSM's experimental 18 and shows that it is optimal.

The proof of Theorem 1.1 is elementary. A left-to-right sweep of the wiring diagram shows that every cycle of $G$ contains an odd number of crossings of a certain type (Lemma 3.2), and an explicit potential function on permutations counts those crossings modulo 2 (Lemma 3.3). Theorem 1.1 is a statement about *every* arrangement, so for $n \equiv \pm 1 \pmod 8$ the obstruction cannot be avoided by choosing the lines cleverly.

**Relation to arrangement graphs.** The *arrangement graph* of a simple arrangement has the intersection points as vertices and the segments between consecutive intersection points on a line as edges [2, 5]. A simple polygon with $\binom{n}{2}$ sides on $n$ lines is the same thing as a Hamiltonian cycle of the arrangement graph that *turns* at every vertex, i.e. never uses two consecutive edges of the same line (Remark 2.4), and by Theorem 1.2 the only candidate for such a cycle is the alternating 2-factor. Bose, Everett and Wismath [2] showed that *some* arrangements of $n$ lines have non-Hamiltonian arrangement graphs, for $n = 6$ and for all odd $n > 6$. Corollary 1.3 is a statement of a different kind: for $n \equiv \pm 1 \pmod 8$ *no* arrangement of $n$ lines or pseudolines has a turning Hamiltonian cycle, whereas for $n = 11$ and $n = 13$ some do. The related notion of an *inducing polygon* goes back to [2], who asked whether every simple arrangement of $n$ lines carries a simple $n$-gon with exactly one side on each line. Scharf and Scherfenberg [12] relaxed the notion to simple polygons whose sides lie on the lines of the arrangement and which use every line (several sides may lie on the same line), i.e. exactly the polygons considered here, and showed that every simple arrangement has such a polygon with $O(n)$ sides; Ackerman, Pinchasi, Scharf and Scherfenberg [1] then answered the original question: an inducing simple $n$-gon always exists. In this language, $M(n)$ is the largest number of sides of an inducing polygon of an arrangement of at most $n$ lines, whereas the cited work is concerned with inducing polygons with *few* sides. We are not aware of earlier work on the maximum number of sides, on the alternating 2-factor, or on the parity of its number of components.

**Verification status.**

- The four constructions of Theorem 1.4 are checked in Lean 4.34.1 (core only, no Mathlib) by kernel evaluation with exact integer arithmetic. The relevant theorems are `mo513798_n7`, `_n9`, `_n11` and `_n13`; the only axiom used is `propext`. See Section 5. They were also rechecked, independently of Lean, in exact rational arithmetic (Section 6).
- Theorems 1.1 and 1.2 and Corollary 1.3 are proved on paper and *not* formally verified. Their conclusions were confirmed by exhaustive computation over all simple arrangements of $n \le 9$ pseudolines, i.e. over all 112 018 190 commutation classes of reduced words of the longest permutation for $n = 9$, and by random tests up to $n = 23$; this is evidence, not a proof (Section 6).

**Organisation.** Section 2 proves Theorem 1.2. Section 3 proves Theorem 1.1. Section 4 gives the constructions and Section 5 their formal certification. Section 6 lists what is not verified, the computational checks, and some open questions.

## 2 Definitions and the reduction

**Definition 2.1.** A *simple polygon with $k$ sides* ($k \ge 3$) is a cyclic sequence of points $P_0, \dots, P_{k-1}$ in the plane with two properties. First, $P_{j-1}, P_j, P_{j+1}$ are never collinear (indices mod $k$), so there are genuine corners. Second, any two non-adjacent sides $[P_i, P_{i+1}]$ and $[P_j, P_{j+1}]$ are disjoint closed segments. Its sides *lie on* a set $\mathcal{L}$ of lines if every side is contained in some line of $\mathcal{L}$. Let $M(n)$ be the largest $k$ for which a simple polygon with $k$ sides has its sides on some set of $n$ lines.

Adjacent sides are not collinear, so they meet only in their common vertex; together with the second condition this says that the boundary is a Jordan curve with exactly $k$ corners.

An arrangement of lines is *simple* if no two lines are parallel and no three are concurrent.

**Definition 2.2** (Alternating 2-factor)**.** Let $\mathcal{A}$ be a simple arrangement of an odd number $n$ of lines. On each line $\ell$, list the $n-1$ intersection points in their order along $\ell$ as $p_1, \dots, p_{n-1}$. The graph $G(\mathcal{A})$ has the $\binom{n}{2}$ intersection points as vertices, and its edges are the pairs

$$\{p_1, p_2\},\ \{p_3, p_4\},\ \dots,\ \{p_{n-2}, p_{n-1}\}$$

on every line.

Every vertex lies on exactly two lines and gets one edge from each of them, so $G$ is 2-regular, i.e. a disjoint union of cycles. Its edges are the first, third, fifth, ... bounded segments of each line.

**Proposition 2.3.** *Let $n$ be odd and let a simple polygon with $k$ sides have its sides on $n$ lines.*

1. *If $k = \binom{n}{2}$, then the arrangement is simple and the polygon's sides are exactly the edges of $G$ (drawn as segments). In particular $G$ is a single cycle.*
2. *If $k \neq \binom{n}{2}$, then $k \leq \binom{n}{2} - 3$.*

*Proof.* Let the sides be $S_j \subset L_j$ ($j \in \mathbb{Z}/k$), with $L_j \neq L_{j+1}$ (consecutive sides are not collinear) and vertices $V_j = L_j \cap L_{j+1}$. The $k$ vertices $P_0, \dots, P_{k-1}$ are distinct: if $P_i = P_j$ with $i \neq j$, then either $j \equiv i \pm 1$ or $j \equiv i \pm 2 \pmod{k}$ and three consecutive vertices, two of which coincide, are collinear, or else the non-adjacent sides $[P_{i-1}, P_i]$ and $[P_j, P_{j+1}]$ share the point $P_i = P_j$; both are excluded by the definition. Hence the pairs $\{L_j, L_{j+1}\}$ are pairwise distinct. These pairs form a set $E'$ of $k$ edges of $K_n$, and the cyclic sequence $(L_j)$ is a closed trail through them. Hence every vertex of $K_n$ has even degree in $E'$.

*Part 2.* $K_n$ is $(n-1)$-regular and $n-1$ is even, so the complement $E(K_n) \setminus E'$ also has all degrees even. If the complement is non-empty it contains a cycle, so it has at least 3 edges.

*Part 1.* Now $E' = E(K_n)$. Every two lines meet, so no two are parallel. The $\binom{n}{2}$ points $V_j$ are distinct, so no three lines are concurrent. Thus the arrangement is simple, and every intersection point is a vertex.

Fix a line $\ell$ with intersection points $p_1, \dots, p_{n-1}$ in order. Each $p_i = \ell \cap m$ is a vertex where a side on $\ell$ meets a side on $m$. So each $p_i$ is an endpoint of exactly one side contained in $\ell$, and there are $(n-1)/2$ such sides. Two sides on $\ell$ are never consecutive, since consecutive sides lie on different lines. Hence they are pairwise disjoint closed segments.

The side with endpoint $p_1$ ends at some $p_m$. If $m > 2$, its relative interior contains $p_2$, which is an endpoint of another side on $\ell$; this contradicts disjointness. So $[p_1, p_2]$ is a side. Repeating the argument on $p_3, \dots$ shows that the sides on $\ell$ are exactly $[p_1, p_2], [p_3, p_4], \dots$, which are the edges of $G$ on $\ell$. The polygon visits all $\binom{n}{2}$ vertices of $G$ along edges of $G$, so $G$ is one cycle. □

*Remark* 2.4. Conversely, if $G$ is a single cycle then it is the boundary of a simple polygon with $\binom{n}{2}$ sides: two edges of $G$ on different lines can only meet at the common intersection point of the two lines, which is then a common vertex, and two edges on the same line are disjoint. The same argument shows that any Hamiltonian cycle of the arrangement graph (Section 1) that turns at every vertex is a simple polygon with $\binom{n}{2}$ sides (two edges of the cycle on the same line cannot share an endpoint, because the cycle turns there, so they are disjoint), so such cycles are exactly the simple $\binom{n}{2}$-gons on the arrangement. We do not need this, because our constructions are verified directly.

**Wiring diagrams.** Rotate the plane so that no line is vertical and all intersection points have distinct $x$-coordinates. Number the lines $1, \dots, n$ from top to bottom far to the left. A vertical line sweeping from left to right sees the vertical order of the lines change exactly at intersection points, each time by swapping the lines at two adjacent heights $i, i+1$. This gives a word $i_1 i_2 \cdots i_N$ with $N = \binom{n}{2}$ in which every pair of lines (*wires*) swaps exactly once, i.e. a reduced word for the longest permutation $w_0$ of $S_n$. This is the wiring diagram of the arrangement [6, 7, 4].

The order of the intersection points along a line equals the order of the corresponding letters in the word. So $G$ depends only on the word, and we define $G$ in the same way for any reduced word of $w_0$. Reduced words of $w_0$ up to commutation of letters $i, j$ with $|i-j| \geq 2$ are the same as simple arrangements of $n$ pseudolines, in the form of wiring diagrams with numbered wires [4]; $G$ is invariant under such commutations, since they do not change the order of the crossings along any wire. All arguments in Section 3 use only the word.

## 3 The parity theorem

Fix $n = 2r+1$ and a reduced word $w = i_1 \cdots i_N$ of $w_0 \in S_n$. Wire $x$ starts at height $x$ (heights $1, \dots, n$ from top to bottom). Letter $i_t$ is the *crossing* $t$: the wires at heights $i_t$ and $i_t+1$ exchange heights. Along each wire, its $n-1$ crossings are numbered $1, \dots, n-1$ in time order. These are its *positions*.

**Definition 3.1.** A crossing is an *$A$-crossing* if it has an even position along both of its wires. Equivalently, both wires have made an odd number of crossings before it. Let $\#A(w)$ be the number of $A$-crossings.

In $G$ every edge joins an odd position to the next even position on one wire. So an $A$-crossing is a vertex both of whose $G$-neighbours occur earlier in time.

**Lemma 3.2.** *Every cycle of $G$ contains an odd number of $A$-crossings. Consequently the number of components of $G$ is congruent to $\#A(w)$ modulo 2.*

*Proof.* Process the crossings in time order. After $t$ steps, call a wire *open* if it has made an odd number of crossings. Its last crossing is then waiting for its $G$-partner along that wire.

Let $G_t$ be the subgraph of $G$ formed by the processed crossings and the edges with both ends processed. Every component of $G_t$ is a cycle of $G$ or a path. Each path has two *dangling ends*, and each dangling end belongs to an open wire; distinct open wires carry distinct dangling ends. A path consisting of a single vertex has both dangling ends at that vertex.

Pairing the two dangling ends of each path gives a perfect matching $\mu_t$ on the set $S_t$ of heights currently occupied by open wires. Consider crossing $t+1$, between wire $u$ at height $i$ and wire $v$ at height $i+1$, and call it $\tau$. Exactly one of the following happens.

(B) *Both wires closed.* $\tau$ has an odd position on both wires. It starts a new path whose dangling ends now sit at heights $i, i+1$, so the pair $(i, i+1)$ is added to $\mu$.

(C) *Exactly one wire open.* Say $u$ is open; the case of $v$ is symmetric. Then $\tau$ extends the path that dangled on $u$. Its new dangling end is on $v$, which now occupies height $i$, the height $u$ had. So $S$ and $\mu$ are unchanged.

(A) *Both wires open.* $\tau$ is an $A$-crossing and joins the dangling ends at heights $i$ and $i+1$. If $\mu(i) = i+1$ this *closes* a cycle of $G$. Otherwise it *joins* two paths: remove $(i, \mu(i))$ and $(i+1, \mu(i+1))$ and add $(\mu(i), \mu(i+1))$.

Wires not involved in $\tau$ do not move.

By induction on $t$ we prove two invariants.

(I1) $\mu_t$ is non-crossing for the order of heights. Case (B) adds a pair of adjacent heights, which cannot interleave with any pair. In case (A) the pairs $(i, \mu(i))$ and $(i+1, \mu(i+1))$ do not cross and $i, i+1$ are adjacent. So they lie either side by side, $\mu(i) < i < i+1 < \mu(i+1)$, or nested, $i < i+1 < \mu(i+1) < \mu(i)$ or its mirror image $\mu(i+1) < \mu(i) < i < i+1$; the other placements of $\mu(i), \mu(i+1)$ relative to $i, i+1$ would make the two pairs cross. A pair crossing the merged pair would cross one of the two old pairs.

(I2) For every pair $(a, b) \in \mu_t$ with $a < b$, the difference $b - a$ is odd. This is true for $(i, i+1)$. Under a side-by-side join, $b - a = (i - a) + 1 + (b - i - 1)$ is a sum of two odd numbers plus 1. Under a nested join, the new difference is (odd)−(odd)−1.

Fix a cycle $C$ of $G$ and let $\Lambda_C(t) = \sum (b - a - 1)/2$, the sum over the pairs of $\mu_t$ whose path lies in $C$. By (I2) this is an integer. We compare $\Lambda_C$ before and after each step that involves $C$:

- a (B)-step adds 0;
- a (C)-step changes nothing;
- closing removes $(i, i+1)$, so it changes $\Lambda_C$ by 0;
- a side-by-side join changes it by $\big((b - a - 1) - (i - a - 1) - (b - i - 2)\big)/2 = 1$;
- a nested join $(i, a) \supset (i+1, b)$ changes it by $-(b - i - 1)$, which is odd by (I2);
- the mirror nested join $(b, i+1) \supset (a, i)$ changes it by $a - i$, again odd.

Steps at crossings not on $C$ do not touch the pairs of $C$: in case (C) the dangling end changes wire but keeps its height, and in cases (A) and (B) only pairs of the paths involved are changed.

The earliest vertex of $C$ has both neighbours later in time, so it is a (B)-vertex and $\Lambda_C = 0$ right after it. Just before $C$ closes, the only pair of $C$ is $(i, i+1)$, so again $\Lambda_C = 0$. Each closing step closes exactly one cycle, so $C$ has exactly one closing step. Therefore $C$ has an even number of joins, and its $A$-crossings (joins plus one closing) are odd in number. Summing over the cycles gives the congruence. □

**Lemma 3.3.** *For every reduced word $w$ of $w_0 \in S_{2r+1}$ we have $\#A(w) \equiv r(r+1)/2 \pmod 2$.*

*Proof.* A wire changes height by $\pm 1$ at each of its crossings. So the number of crossings it has made so far is congruent modulo 2 to its current height minus its label.

Consider a crossing at heights $(i, i+1)$ of wires $u$ and $v$. They have not crossed before, so their initial order persists and $u < v$. The crossing is an $A$-crossing iff $i - u$ and $i + 1 - v$ are both odd. If $u + v$ is even, then $i - u$ and $i + 1 - v$ have different parities and this is impossible. If $u + v$ is odd, they have the same parity and the crossing is an $A$-crossing iff $i - u$ is odd. In both cases, writing $[\,\cdot\,] \in \{0, 1\}$ for the truth value of a statement,

$$[A\text{-crossing}] = [u + v \text{ odd}] \cdot [i + u \text{ odd}]. \tag{3.1}$$

*A potential function.* A state of the sweep is a permutation $\pi$, where $\pi(h)$ is the wire at height $h$. Put

$$\Phi(\pi) = \sum_{h=1}^{n} [\pi(h) \text{ odd}] \Big\lfloor \frac{h}{2} \Big\rfloor \;+\; \#\{\{x, y\} :\; x < y,\; x \text{ odd},\; y \text{ even},\; x \text{ and } y \text{ have already crossed}\}.$$

We claim that every crossing changes $\Phi$ by an amount congruent modulo 2 to its $A$-status. Let the crossing be as above. In the sum only the terms at heights $i$ and $i+1$ change: wire $u$ moves from height $i$ to $i+1$ and wire $v$ from $i+1$ to $i$, so the sum changes by

$$([u \text{ odd}] - [v \text{ odd}])\Big(\Big\lfloor \frac{i+1}{2} \Big\rfloor - \Big\lfloor \frac{i}{2} \Big\rfloor\Big) \equiv [u+v \text{ odd}] \cdot [i \text{ odd}] \pmod 2.$$

The second term increases by exactly $[u \text{ odd}][v \text{ even}] = [u+v \text{ odd}] \cdot [u \text{ odd}]$, because the pair $\{u, v\}$ becomes crossed and no other pair changes its status. Adding, the change of $\Phi$ is congruent to $[u+v \text{ odd}]([i \text{ odd}] + [u \text{ odd}]) \equiv [u+v \text{ odd}] \cdot [i+u \text{ odd}]$, which is the $A$-status by (3.1).

Summing over the $N$ crossings of $w$ gives $\#A(w) \equiv \Phi(w_0) - \Phi(\mathrm{id}) \pmod 2$, where id is the initial state and $w_0$, with $w_0(h) = n+1-h$, the final one. Since $n+1$ is even, $w_0(h)$ is odd iff $h$ is odd, so the sums in $\Phi(\mathrm{id})$ and $\Phi(w_0)$ agree. Initially no pair has crossed and finally every pair has. Hence

$$\#A(w) \equiv \#\{\{x,y\} : x < y,\ x \text{ odd},\ y \text{ even}\} = \sum_{m=1}^{r} \#\{x \text{ odd} : x < 2m\} = \sum_{m=1}^{r} m = \frac{r(r+1)}{2}. \quad \square$$

*Remark* 3.4. Lemma 3.3 can also be derived from the theorem of Matsumoto and Tits [9, 13], which says that any two reduced words of $w_0$ are connected by commutation moves (exchanging adjacent letters $i, j$ with $|i-j| \geq 2$) and braid moves $(i, i+1, i \leftrightarrow i+1, i, i+1)$. A commutation move changes neither the wires nor the heights of any crossing. A braid move involves three wires $x < y < z$; each of the crossings $xy, xz, yz$ keeps its wires and its upper height changes by exactly 1, so by (3.1) the move flips the $A$-status of exactly those pairs among $x, y, z$ whose labels have odd sum, and there are 0 or 2 such pairs. Hence $\#A$ mod 2 is the same for all reduced words of $w_0$, and the bubble-sort word $(s_1 \cdots s_{n-1})(s_1 \cdots s_{n-2}) \cdots (s_1)$, in which the crossing of $u < v$ has position $v-1$ on $u$ and position $u$ on $v$, gives the value $\#\{u \text{ even},\ v \text{ odd},\ u < v\} = r(r+1)/2$. The potential $\Phi$ makes this detour unnecessary.

*Proof of Theorem 1.1.* Combine Lemmas 3.2 and 3.3, and note that $(n^2-1)/8 = r(r+1)/2$. $\square$

*Proof of Corollary 1.3.* For $n = 2r+1$, the number $r(r+1)/2$ is even iff $r \equiv 0, 3 \pmod 4$, i.e. iff $n \equiv \pm 1 \pmod 8$. In that case $G$ has an even number of components, so it is not a single cycle. By Proposition 2.3(1), no simple $\binom{n}{2}$-gon lies on $n$ lines, and by Proposition 2.3(2) every such polygon has at most $\binom{n}{2} - 3$ sides. $\square$

*Remark* 3.5. For $n \equiv \pm 3 \pmod 8$, Theorem 1.1 only says that $G$ has an odd number of components. It does not guarantee a single cycle; for example, both 1 and 3 components occur for $n = 5$ (Section 6).

## 4 Constructions

We write the line $\ell_m$ as $a_m x + b_m y = c_m$ with integer coefficients. Each polygon is given by the cyclic sequence of its vertices, where the vertex "$p/q$" is $\ell_p \cap \ell_q$. Consecutive vertices share a line, which carries the side between them. All data, including vertex coordinates as exact rationals, are in `data/polyN.json` and `lean/Main.lean`.

| $n$ | lines $(a_m, b_m, c_m)$, $m = 1, \dots, n$ |
|---|---|
| 7 | $(-6, 17, -74)$, $(0, -19, -95)$, $(-19, 14, -98)$, $(4, -7, 8)$, $(-19, 13, -44)$, $(8, 11, 41)$, $(-6, 2, -41)$ |
| 9 | $(-22, 6, -118)$, $(-14, -23, 103)$, $(18, -2, 91)$, $(11, -6, -43)$, $(-24, 1, -136)$, $(27, 23, 49)$, $(-3, 8, -149)$, $(14, -2, -14)$, $(16, 21, -33)$ |
| 11 | $(-39, 31, -51)$, $(1, 50, 7)$, $(-22, 45, -264)$, $(12, -49, -212)$, $(-10, -49, -25)$, $(-16, 48, 31)$, $(48, -14, 68)$, $(-19, 46, -80)$, $(-50, 3, 89)$, $(17, 47, 93)$, $(15, 48, 37)$ |
| 13 | $(5, -19, 255)$, $(17, 11, 229)$, $(-4, 20, -1118)$, $(14, -14, 106)$, $(19, 4, 126)$, $(3, -20, -505)$, $(-6, 19, -402)$, $(10, 17, 78)$, $(20, -3, -68)$, $(-1, 20, 240)$, $(18, -8, 29)$, $(7, -19, -397)$, $(-20, -3, -59)$ |

Table 1: The certified arrangements.

**$n = 7$, an 18-gon.** The vertex cycle is

$$1/2,\ 1/6,\ 6/7,\ 1/7,\ 1/3,\ 3/5,\ 1/5,\ 1/4,\ 4/5,\ 5/6,\ 2/6,\ 2/5,\ 5/7,\ 2/7,\ 2/3,\ 3/7,\ 4/7,\ 2/4.$$

It uses every intersection point except the three points of the triangle of lines $\{3, 4, 6\}$. Lines $3, 4, 6$ each carry 2 sides and the other lines carry 3. The line pairs used form $K_7$ minus a triangle, which is the extremal case of Proposition 2.3(2).

**$n = 9$, a 33-gon.** This uses all intersection points except those of the triangle of lines $\{1, 2, 9\}$; the cycle is listed in Section A.

**$n = 11$ and $n = 13$.** Here the polygon is the alternating 2-factor $G$ itself, which is a single cycle of length 55, respectively 78; it is listed in Section A. Each line carries $(n - 1)/2$ sides.

**How they were found.** For $n = 11, 13$ we ran a simulated-annealing search over random real arrangements, minimising the number of components of $G$. Theorem 1.1 predicts an odd number for these $n$, and the search eventually reaches one component. The resulting coefficients were rounded to small integers and re-checked with exact rational arithmetic.

For $n = 7, 9$ we sampled random integer arrangements and, for each triangle $\{a, b, c\}$ of lines, formed the graph that pairs consecutive remaining intersection points on every line, leaving out $\ell_a \cap \ell_b$, $\ell_b \cap \ell_c$ and $\ell_a \cap \ell_c$. We kept the arrangement when this graph was a single cycle bounding a simple polygon.

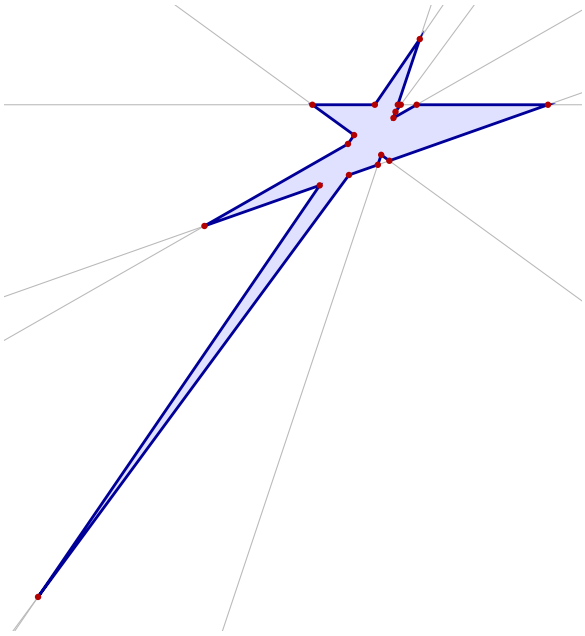
The search method plays no role in correctness. Every polygon is verified directly, as described in Section 5.

*Proof of Theorem 1.4.* The four polygons are simple and have their sides on the listed lines; this is certified by the Lean theorems of Section 5. The lower bounds $M(n) \geq 18, 33, 55, 78$ follow.

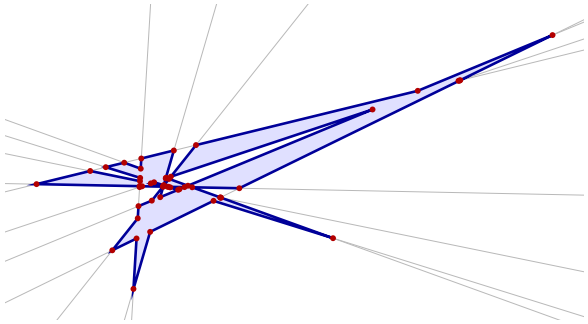
For the matching upper bounds: $M(11) \leq 55$ and $M(13) \leq 78$ hold by the closed-trail bound $n\lfloor (n-1)/2 \rfloor$. $M(7) \leq 18$ and $M(9) \leq 33$ hold by Corollary 1.3. □

## 5 Formal verification of the constructions

The file `lean/Main.lean` is self-contained and was checked with Lean 4.34.1 using only the core library and Std; Mathlib is not used. The check took 37–53 s of wall time and about 2 GB of memory on the authors' machine; an independent re-check with the same Lean version on another machine took 19 s and 1.99 GB and produced the same output. The main statement is the following.



18 sides on 7 lines



55 sides on 11 lines

Figure 1: Two of the certified polygons, drawn to scale. The grey lines are the arrangement, clipped to a box around the polygon. Several sides are very short, so some corners are not resolved at this scale; exact coordinates are in the data files.

```
theorem mo513798_n11 :
    ∃ (L : List Ln) (P : List Pt),
      L.length = 11 ∧ P.length = 55 ∧ genuineLines L = true ∧
      isSimplePolygon P = true ∧ sidesOnLines L P = true :=
  ⟨lines11, poly11, by decide +kernel, by decide +kernel, by decide +kernel,
    by decide +kernel, by decide +kernel⟩
```

The theorems `mo513798_n13`, `mo513798_n7` and `mo513798_n9` have the same form, with $(n, k) = (13, 78)$, $(7, 18)$ and $(9, 33)$. All Boolean checks are discharged by `decide +kernel`, i.e. by evaluation in the kernel. No `sorry`, `axiom`, `admit` or `native_decide` occurs. The verbatim output of `#print axioms` (`lean/lean_output.txt`) is:

```
'mo513798_n11' depends on axioms: [propext]
'mo513798_n13' depends on axioms: [propext]
'mo513798_n7' depends on axioms: [propext]
'mo513798_n9' depends on axioms: [propext]
```

`propext` is a standard axiom of Lean. It enters through core decidability instances.

**Definitions used by the statement** (verbatim, doc-comments removed):

```
structure Pt where
  x : Int
  y : Int
  w : Int
deriving Repr
```

```
structure Ln where
  a : Int
  b : Int
  c : Int
deriving Repr

def onLine (l : Ln) (p : Pt) : Bool := l.a * p.x + l.b * p.y == l.c * p.w

def orient (p q r : Pt) : Int :=
  p.x * (q.y * r.w - q.w * r.y) - p.y * (q.x * r.w - q.w * r.x) + p.w * (q.x * r.y - q.y * r.x)

def fle (u uw v vw : Int) : Bool := decide (u * vw ≤ v * uw)

def inBox (p q r : Pt) : Bool :=
  ((fle p.x p.w r.x r.w && fle r.x r.w q.x q.w) || (fle q.x q.w r.x r.w && fle r.x r.w p.x p.w))
    &&
  ((fle p.y p.w r.y r.w && fle r.y r.w q.y q.w) || (fle q.y q.w r.y r.w && fle r.y r.w p.y p.w))

def segInter (a b c d : Pt) : Bool :=
  let d1 := orient c d a
  let d2 := orient c d b
  let d3 := orient a b c
  let d4 := orient a b d
  ((decide (d1 > 0) && decide (d2 < 0)) || (decide (d1 < 0) && decide (d2 > 0))) &&
    ((decide (d3 > 0) && decide (d4 < 0)) || (decide (d3 < 0) && decide (d4 > 0))) ||
  (d1 == 0 && inBox c d a) || (d2 == 0 && inBox c d b) ||
  (d3 == 0 && inBox a b c) || (d4 == 0 && inBox a b d)

def vtx (P : List Pt) (j : Nat) : Pt := P.getD (j % P.length) ⟨0, 0, 1⟩

def isSimplePolygon (P : List Pt) : Bool :=
  let k := P.length
  decide (3 ≤ k) &&
  P.all (fun p => decide (p.w > 0)) &&
  (List.range k).all (fun j => orient (vtx P j) (vtx P (j + 1)) (vtx P (j + 2)) != 0) &&
  (List.range k).all (fun i => (List.range k).all (fun j =>
    if decide (i + 2 ≤ j) && decide (j + 2 ≤ k + i) then
      !(segInter (vtx P i) (vtx P (i + 1)) (vtx P j) (vtx P (j + 1)))
    else true))

def sidesOnLines (L : List Ln) (P : List Pt) : Bool :=
  (List.range P.length).all (fun j =>
    L.any (fun l => onLine l (vtx P j) && onLine l (vtx P (j + 1))))

def genuineLines (L : List Ln) : Bool := L.all (fun l => l.a != 0 || l.b != 0)
```

**Statement audit.** We explain why the Lean statement means exactly "there are $n$ lines and a simple polygon with $k$ sides, every side lying on one of the lines".

- *Points.* A point is an integer triple $(x, y, w)$; it represents $(x/w, y/w) \in \mathbb{Q}^2$. `isSimplePolygon` checks $w > 0$ for every vertex.
- *Lines.* `onLine` tests $ax + by = cw$, which is $a\frac{x}{w} + b\frac{y}{w} = c$ multiplied by $w > 0$. `genuineLines` excludes the degenerate equation $0x + 0y = c$, which would contain every point when $c = 0$.
- *Orientation.* `orient` is the $3 \times 3$ determinant of the homogeneous coordinates. It equals

$p_w q_w r_w \cdot ((q-p) \times (r-p))$ for the affine points. Since the weights are positive, its sign is the orientation of $(p, q, r)$, and it vanishes iff the points are collinear.

- *Segment intersection.* `segInter` is the standard exact test for closed segments [3, §33.1]: a proper crossing, or an endpoint of one segment lying on the other. The box test `inBox` is used only when an orientation is 0, i.e. for collinear points. In that case "lies in the bounding box" is equivalent to "lies on the segment". `fle` compares fractions with positive denominators.

- *Simple polygon.* Vertex $j$ is `P[j mod k]. isSimplePolygon` requires:

  - $k \geq 3$;
  - every triple of consecutive vertices is non-collinear;
  - for $i < j$, the sides $i$ and $j$ are disjoint whenever $i + 2 \leq j$ and $j + 2 \leq k + i$.

  The last condition ranges over exactly the non-adjacent pairs of sides; the adjacent pairs are $j = i + 1$ and $(i, j) = (0, k - 1)$. Non-collinearity of consecutive triples makes consecutive vertices distinct and makes adjacent sides meet only in their common vertex. It also forbids two collinear consecutive sides, so the polygon has exactly $k$ genuine sides. These conditions say exactly that the boundary is a Jordan polygon with $k$ sides, as in the definition in Section 2.

- *Sides on lines.* `sidesOnLines` asks that both endpoints of each side lie on a common line of the list. Then the whole side lies on that line. The two length conditions fix $n$ and $k$. If two listed lines coincided, the statement would only become stronger. In fact the listed lines are pairwise distinct and in general position; this was checked separately in exact rational arithmetic.

- *Checker sanity.* `Main.lean` also contains `example`s showing that the checker behaves as intended:

  - it accepts a unit square;
  - it rejects a bow-tie, three collinear consecutive vertices, and a polygon that touches itself;
  - collinear segments are treated correctly whether they overlap or are disjoint.

**Reproduction.** The file has no `import` lines, so `lean lean/Main.lean` with Lean 4.34.1 checks it and prints the four `#print axioms` lines above. Independently of Lean, the script `scripts/verify_constructions.py` rechecks simplicity, general position and the structure of the four polygons with exact rational arithmetic (Section 6).

## 6 What is not verified; computational checks; open questions

**Not formally verified.** The following are proved on paper only.

- Proposition 2.3, including the reduction from line arrangements to wiring diagrams.
- Lemma 3.2, the sweep argument.
- Lemma 3.3, the potential-function computation.

- Hence Theorem 1.1, Corollary 1.3, and the upper bounds $M(7) \leq 18$ and $M(9) \leq 33$.

For each fixed $n$, the one-step identity for $\Phi$ in the proof of Lemma 3.3 is a finite statement about the $n!$ sweep states, and Lemma 3.2 is a finite statement about the commutation classes of reduced words of $w_0$. Both are natural targets for a kernel-checked Lean proof, but we have not produced one. An earlier attempt to certify Lemma 3.3 for $n = 7$ in Lean, by a kernel-evaluated breadth-first search over the sweep states of all 5040 permutations, ran out of kernel memory; we chose not to fall back on `native_decide`.

**Computational checks.** These are evidence, not proofs. The scripts are in the directory `scripts/` next to this paper; each was run on October 8, 2026.

- `enumerate_classes.c` enumerates every commutation class of reduced words of $w_0$, i.e. every simple arrangement of $n$ pseudolines as a wiring diagram with numbered wires, by generating the lexicographically least word of each class. For each class it builds $G$ from the order of the crossings along each wire, counts the components, and counts the $A$-crossings on each cycle. For $n = 5, 7, 9$ it visits all 62, 24 698 and 112 018 190 classes; these counts agree with OEIS A006245 [10]. In every class, $\#A \equiv r(r+1)/2 \pmod 2$ and every cycle of $G$ contains an odd number of $A$-crossings. The numbers of components are distributed as follows.

| $n$ | components of $G$ (number of classes) |
|---|---|
| 3 | 1 (2) |
| 5 | 1 (22), 3 (40) |
| 7 | 2 (7 870), 4 (10 220), 6 (6 608) |
| 9 | 2 (6 360 134), 4 (35 038 062), 6 (39 168 090), 8 (23 868 000), 10 (7 583 904) |

  In particular $G$ is never a single cycle for $n = 7$ or $n = 9$. Since every simple arrangement of lines gives a reduced word of $w_0$, this confirms $M(7) \leq 18$ and $M(9) \leq 33$ independently of Section 3, though it still relies on Proposition 2.3. The pure Python version `enumerate_classes.py` does the same for $n \leq 7$.

- `potential_check.py` verifies the one-step identity for $\Phi$ in the proof of Lemma 3.3 for every permutation and every ascent, for $n \leq 8$.

- `parity_random.py` checks the parity of the number of components of $G$ and the per-cycle statement of Lemma 3.2 on random reduced words of $w_0$, for every odd $n \leq 23$ (3000 words for $n \leq 13$, 500 words for $15 \leq n \leq 23$). No violation occurred. The smallest numbers of components seen were $6, 8, 9, 9, 12$ for $n = 15, 17, 19, 21, 23$.

- `verify_constructions.py` rechecks the four polygons of Section 4 with exact rational arithmetic, independently of Lean: the vertices in `lean/Main.lean` are the intersection points named in `data/polyN.json`, the polygons are simple in the sense of Section 2, the arrangements are in general position, the polygons for $n = 11, 13$ coincide with $G$, and the polygons for $n = 7, 9$ use all intersection points except those of one triangle.

**Open questions.**

1. Is $M(n) = \binom{n}{2}$ for every $n \equiv \pm 3 \pmod 8$? We confirmed this for $n = 3, 5, 11, 13$. Brief simulated-annealing searches for $n = 19$ and $n = 21$ (four runs of $10^6$ steps each) reached 3 components but not 1.

2. Is $M(n) = \binom{n}{2} - 3$ for every $n \equiv \pm 1 \pmod 8$? We confirmed this for $n = 7$ and $n = 9$. The extremal polygons of Proposition 2.3(2) use all pairs of lines except one triangle; it is not clear that such a polygon exists for every $n \equiv \pm 1 \pmod 8$.

3. What is $M(n)$ for even $n$? The closed-trail bound is $n(n-2)/2$, and DSM's 24-gon on 8 lines attains it for $n = 8$. We have not studied even $n$; note that Definition 2.2 needs $n - 1$ to be even.

4. Which numbers of components does $G$ have, for a given $n$, beyond the parity constraint? By the table above the minimum is 1 for $n = 3, 5$ and 2 for $n = 7, 9$; by Section 4 it is 1 for $n = 11, 13$. Is the minimum always 1 or 2 according to the parity?

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# A Vertex cycles of the larger constructions

The notation "$p/q$" means $\ell_p \cap \ell_q$, with lines numbered as in Table 1. Each list is the cyclic order of the vertices.

**$n = 9$ (33 vertices; lines $1, 2, 9$ form the omitted triangle).** 1/3, 1/6, 3/6, 3/9, 5/9, 5/6, 6/9, 7/9, 2/7, 2/6, 6/7, 5/7, 2/5, 2/3, 3/7, 1/7, 1/8, 7/8, 4/7, 2/4, 2/8, 8/9, 4/9, 4/6, 6/8, 4/8, 4/5, 5/8, 3/8, 3/4, 1/4, 1/5, 3/5.

**$n = 11$ (55 vertices).** 1/2, 1/8, 8/9, 1/9, 1/3, 3/9, 7/9, 3/7, 3/11, 10/11, 5/10, 3/5, 3/10, 2/10, 2/3, 3/4, 4/6, 3/6, 3/8, 4/8, 1/4, 1/6, 6/8, 8/10, 1/10, 1/11, 2/11, 2/8, 5/8, 5/11, 8/11, 7/8, 2/7, 2/5, 5/7, 1/7, 1/5, 5/6, 6/11, 7/11, 6/7, 6/10, 7/10, 4/7, 4/9, 9/10, 4/10, 4/11, 9/11, 5/9, 4/5, 2/4, 2/9, 6/9, 2/6.

**$n = 13$ (78 vertices).** 1/2, 1/5, 4/5, 4/8, 5/8, 5/11, 8/11, 8/13, 11/13, 4/11, 4/13, 1/13, 1/11, 9/11, 1/9, 1/4, 4/9, 8/9, 8/10, 9/10, 9/12, 8/12, 6/8, 6/10, 10/12, 1/12, 1/3, 3/12, 7/12, 3/7, 3/4, 4/7, 7/11, 3/11, 3/9, 7/9, 7/13, 3/13, 3/5, 5/7, 2/7, 2/3, 3/8, 7/8, 7/10, 3/10, 3/6, 1/6, 1/7, 6/7, 4/6, 4/12, 6/12, 6/11, 11/12, 5/12, 5/6, 6/9, 5/9, 5/13, 6/13, 2/6, 2/9, 9/13, 2/13, 2/12, 12/13, 10/13, 5/10, 2/5, 2/10, 10/11, 2/11, 2/4, 4/10, 1/10, 1/8, 2/8.